

Docker Day 01 - Fundamentals

Objective

Understand Docker fundamentals and containerization.

Topics

Docker, Containers, Images, Docker Hub, Docker Engine, Docker Daemon, Lifecycle.

Commands

docker --version, docker version, docker info, docker images, docker ps, docker ps -a, docker run hello-world, docker stop, docker start, docker restart, docker rm, docker rmi.

Architecture

Docker Client -> Docker Engine -> Docker Daemon -> Images/Containers -> Docker Hub.

Lifecycle

Dockerfile -> Image -> Container -> Stop -> Start -> Remove.

Production Workflow

Developer -> Build -> Registry -> Server -> Run Container.

Best Practices

Use official images, pin versions, keep images small, remove unused resources, avoid secrets in images.

Interview Questions

What is Docker? Docker vs VM? Image vs Container? What is Docker Daemon?